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Hybrid learning-based fault prediction and cascading failure mitigation in multi-network energy systems

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This paper introduces a novel approach for managing fault propagation in interconnected energy networks comprising electric, gas, and heating systems. As energy infrastructures become increasingly integrated, the risk of cascading failures across these networks grows, making it critical to develop robust models for predicting and mitigating fault propagation. To tackle the complexity of fault propagation in interconnected energy systems, we develop a novel AI-based management architecture that couples adversarial learning mechanisms with graph-structured predictive models. Specifically, a generative network is employed to synthesize plausible fault evolution patterns from historical records, while a graph-based neural architecture captures the spatiotemporal correlations among the subsystems. Furthermore, a robust optimization scheme under distributional uncertainty is incorporated to devise adaptive recovery strategies, enhancing the resilience and reliability of system restoration processes. The proposed model is tested using a synthetic case study based on the IEEE 123-bus electric network, the Belgian gas transmission system, and a standard heating network. The results demonstrate the effectiveness of the model in accurately predicting fault propagation and optimizing recovery strategies, significantly reducing recovery time and minimizing the impact of cascading failures. The primary innovations presented in this study are the development of an integrated fault propagation framework spanning multiple energy networks, the pioneering use of adversarial and graph-based learning techniques for fault trajectory prediction, and the incorporation of distributionally robust optimization to strengthen recovery planning. Collectively, these advancements contribute to a deeper understanding of fault dynamics in interconnected infrastructures and propose a scalable pathway for enhancing system resilience in modern energy networks.

Keywords Fault propagation, Interconnected energy networks, Generative adversarial networks (GANs), Distributionally robust optimization (DRO), Multi-network systems, Cascading failures, Graph neural networks (GNNs)

In recent years, the increasing complexity and interdependency of modern energy networks, specifically electric, gas, and heating systems, have created both unprecedented opportunities and significant challenges. These interconnected energy networks are becoming integral to the infrastructure of cities and industrial areas, supporting large-scale energy consumption while maintaining operational efficiency[1,2]. However, the integrated nature of these systems also makes them vulnerable to cascading failures. A fault in one network, such as a power outage in the electric grid, can propagate rapidly to the gas or heating networks, creating a multi-network fault propagation scenario[3]. As the complexity of these systems continues to grow, so does the need for robust methods to predict, mitigate, and recover from faults. The motivation behind this research lies in addressing the increasingly evident risks posed by multi-network interactions during fault propagation. Modern urban areas rely on these interconnected systems to provide reliable energy for residential, commercial, and industrial activities. Yet, the interdependence that enhances their efficiency also increases the likelihood of systemic failures, where a single disturbance in one network can have wide-reaching consequences across others[4,5].

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This is especially true during high-impact, low-probability events such as extreme weather conditions, cyber-attacks, or equipment failures, where the failure of one network can propagate through others, exacerbating the overall disruption. Existing methods for fault management, which tend to treat these networks in isolation, are inadequate for addressing the complex interactions and cascading effects that occur when faults propagate across multiple energy networks.

With the increasing deployment of AI algorithms for forecasting, optimization, and control in modern PV grids, the system's cyber-physical attack surface has expanded substantially. Adversaries can now target not only physical components but also AI-driven decision pipelines, such as neural network-based controllers or data aggregation modules, making the grid vulnerable to stealthy, adaptive, and coordinated cyberattacks. Existing defense strategies often rely on static thresholds or fixed-rule intrusion detection systems, which are insufficient in dealing with intelligent, dynamic threats that evolve over time or mimic legitimate behavior.

To address these challenges, our proposed defense architecture introduces a game-theoretic framework that explicitly models the strategic interaction between attackers and defenders, allowing the system to anticipate adversarial moves and respond optimally. Moreover, we incorporate reinforcement learning-based adaptation, enabling the system to continuously update its defensive policies in response to evolving threat patterns and environmental feedback. This dual-layered design fills a practical gap in current research: while many existing approaches treat cyber-defense as either a classification or anomaly detection task, few consider the need for real-time, sequential decision-making under adversarial uncertainty. By combining predictive modeling, strategic planning, and learning-based adaptation, our framework offers a proactive and reactive cyber-resilience solution tailored for AI-enabled PV grid infrastructures.

Figure 1 illustrates the proposed multi-layered cyber-defense framework for AI-enabled PV grids. The flowchart presents a sequential interaction of five functional layers: Threat Modeling, Game-Theoretic Defense, Behavior Analysis, Risk Assessment, and Defensive Actions. The process begins with the identification of cyberattack vectors, which are decomposed into attack surfaces through threat modeling. This information feeds into a game-theoretic engine that simulates adversarial behavior and computes optimal defensive strategies. Behavior analysis continuously monitors system responses and refines anomaly detection. The insights are then evaluated through dynamic risk assessment to quantify potential service disruptions. Finally, the framework triggers adaptive defensive actions that mitigate emerging threats. The diagram emphasizes feedback loops between layers to enable real-time learning and closed-loop resilience against evolving cyber-physical attacks. While prior studies have recognized the risks of cascading failures in interdependent energy systems, most existing approaches focus on isolated network modeling or assume static fault scenarios that overlook dynamic,

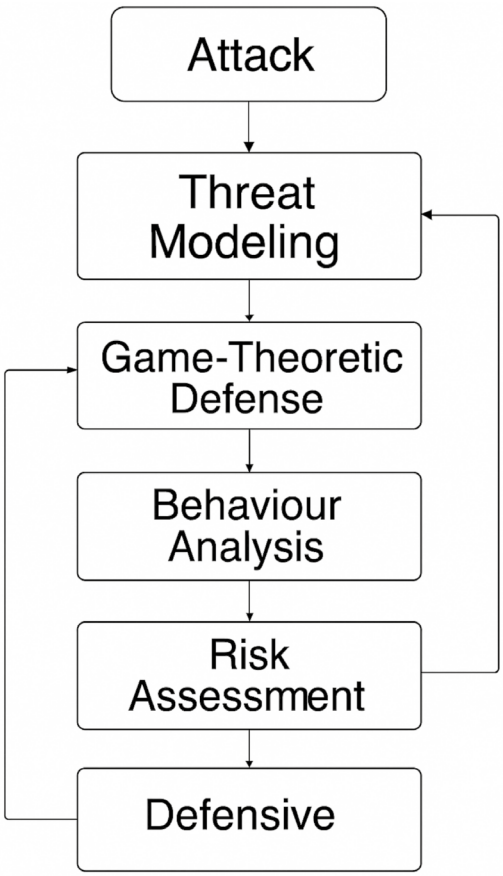


Fig. 1. Layered cyber-defense architecture for AI-enabled PV grids.

cross-domain interactions. Traditional fault detection and mitigation strategies often rely on deterministic rules or local optimization without accounting for the spatiotemporal propagation of disturbances across electric, gas, and heating infrastructures. Moreover, many methods treat uncertainty in a simplified or bounded fashion, failing to capture the distributional variability and strategic nature of cyber-physical disruptions. These limitations reduce the robustness and applicability of current frameworks, especially in the context of increasingly integrated and AI-driven energy ecosystems. To address these gaps, this work proposes a unified architecture that integrates generative adversarial modeling, graph-based learning, and distributionally robust optimization, enabling adaptive, data-driven decision-making for fault prediction and recovery across coupled multi-energy systems.

This paper proposes a novel approach to understanding and mitigating fault propagation in interconnected electric, gas, and heating networks. The focus of this work is to develop a comprehensive modeling framework that can predict fault propagation across these networks and optimize recovery strategies. This study adopts state-of-the-art machine learning paradigms, including adversarial generation models and graph-based relational learning, alongside advanced optimization strategies, to propose a novel framework for fault management capable of addressing the intrinsic uncertainties and intricate interdependencies within multi-energy network infrastructures. The key research objective of this paper is to build an AI-driven fault propagation model that can accurately simulate the dynamic interactions between the electric, gas, and heating networks during fault scenarios. This model not only predicts the path of fault propagation but also aids in optimizing recovery strategies by accounting for the interdependencies among the networks. The methodology developed in this research integrates GANs for generating realistic fault scenarios and Graph Neural Networks (GNNs) for capturing the spatial-temporal fault dynamics across interconnected networks. Moreover, employing a distributionally robust optimization framework enhances the adaptability of recovery strategies against uncertainties in fault dynamics, thereby equipping the proposed model to effectively manage practical fault conditions with superior precision and operational reliability.

The proposed modeling architecture introduces an original design tailored to address the complexities of fault propagation across interconnected energy networks. Initially, a generative model synthesizes plausible fault evolution scenarios derived from historical records and established fault patterns. These generated scenarios are fed into the GNN, which learns the patterns of fault propagation across the networks. The GNN is trained to capture the spatiotemporal dependencies among network nodes, enabling the simulation of fault dissemination trajectories across the electricity, gas, and thermal infrastructures over temporal horizons. Following the characterization of fault evolution patterns, a distributionally robust optimization framework is employed to devise recovery strategies that maintain system resilience even under extreme fault conditions. The overall framework is designed to provide a comprehensive solution for managing faults in interconnected energy systems, accounting for the dynamic, uncertain, and interdependent nature of these networks. This paper makes 3 key contributions to the field of fault management in interconnected energy networks:

1. Novel Application of Generative Adversarial Networks (GANs) and GNNs for Fault Propagation Modeling: This paper introduces the application of GANs to generate realistic fault propagation scenarios across interconnected electric, gas, and heating networks. These scenarios are used to train a GNN, which models the dynamic interactions between the networks and predicts the spatial-temporal propagation of faults. The integration of GANs and GNNs in this context is a novel contribution that enhances the accuracy and effectiveness of fault prediction.
2. Holistic Modeling of Multi-Network Interactions: This approach offers a more realistic and effective solution to managing faults in interconnected energy systems. In contrast to traditional studies that predominantly address individual energy systems in isolation, this work introduces an integrated modeling framework designed to characterize the dynamic couplings among electricity, gas, and thermal networks. The proposed fault propagation framework explicitly accounts for the mutual dependencies across these infrastructures, offering deeper insights into the mechanisms by which disturbances in one network can influence and trigger cascading effects in the others.
3. Robust Optimization for Fault Recovery: The application of DRO in this paper ensures that recovery strategies are robust to uncertainties in fault propagation. By considering the worst-case distribution of fault scenarios within an uncertainty set, the DRO framework ensures that the proposed recovery strategies are resilient to unexpected faults. This is particularly important in real-world applications, where fault scenarios are often unpredictable and can deviate significantly from historical patterns.

In conclusion, this paper addresses a critical gap in fault management for interconnected energy networks by introducing a novel, AI-driven framework that combines GANs, GNNs, and DRO. By modeling fault propagation across electric, gas, and heating networks and optimizing recovery strategies under uncertainty, the proposed approach offers a powerful solution for managing cascading failures in modern energy systems. This research contributes to the advancement of fault management techniques and provides a pathway toward more resilient, reliable, and efficient energy infrastructure.

Literature review

Effectively addressing faults within interconnected energy infrastructures has become a pressing challenge amid the rising structural complexity of contemporary energy systems. Networks that integrate electricity, gas, and thermal services are fundamental to supporting urban activities and industrial operations. However, the interdependence between these systems means that a fault in one network can propagate to others, leading to cascading failures and widespread disruptions[6,7]. The literature on fault management in interconnected energy systems is vast, spanning multiple disciplines including electrical engineering, gas network optimization, heat

transfer, AI, and robust optimization. This section reviews the key contributions in these areas, focusing on fault propagation modeling, AI-based approaches, and robust optimization strategies that inform the development of the comprehensive framework proposed in this paper. The study of fault propagation in energy networks has a long history, with initial work focusing on the independent analysis of electrical systems. Early fault models in power systems were based on deterministic approaches, which aimed to detect, isolate, and mitigate faults using predefined rules[8,9]. For example, traditional protection mechanisms such as relays and circuit breakers were designed to respond to overcurrent or voltage deviations within the power grid[10]. However, these methods are often limited in their ability to handle more complex, interconnected systems[11]. Recent studies have shifted toward more comprehensive models that account for interactions between multiple networks, including electric, gas, and heating systems. Researchers have highlighted the importance of considering interdependencies between these networks, noting that faults in one system can have cascading effects on others. For instance, research[12] studied the interactions between electric and gas networks, demonstrating how power outages in the electric grid can disrupt gas pipelines, which in turn affects electricity generation in gas-fired plants. Similarly, reference[13] explored the coupling between the electric and heating networks, showing that electricity shortages can significantly reduce heating efficiency in urban areas during cold weather events. The challenge of modeling fault propagation across interconnected networks has led to the development of several novel approaches. For instance, research[14] proposed a hybrid analytical-simulation model to examine cascading failures in power and gas systems. Their study emphasized the need for integrating both systems in a unified framework to understand the full scope of fault propagation. Nevertheless, these models are typically static and focus on specific types of interactions, limiting their ability to capture the dynamic, time-varying nature of fault propagation across multiple energy networks.

Artificial intelligence has been increasingly applied to fault detection and prediction in energy networks, with methods such as machine learning, deep learning, and reinforcement learning gaining prominence. Traditional machine learning techniques, such as support vector machines (SVMs) and decision trees, have been used to classify and predict faults based on historical data[15,16]. However, these methods often require large datasets and may struggle to generalize beyond known fault types. In recent years, deep learning techniques such as convolutional neural networks (CNNs) and recurrent neural networks (RNNs) have shown promise in handling the complexities of fault prediction in energy systems[17,18]. For instance, reference[19] developed a deep learning-based method to predict cascading failures in the electric grid, demonstrating superior accuracy compared to traditional methods. Their approach utilized a CNN to detect patterns in the data and predict how faults might propagate through the grid. Similarly, GNNs have been applied to fault prediction in networked systems due to their ability to model relationships between nodes in a graph-like structure, which is naturally suited to energy systems. Reference[20] proposed a GNN-based method for predicting fault propagation in power systems, capturing the spatial dependencies between nodes in the network. Generative models, particularly GANs, have also emerged as powerful tools for generating synthetic data that closely mimics real-world scenarios. GANs have been widely used in areas such as image generation and natural language processing, but their application to fault prediction in energy systems is relatively new. Reference[21] introduced a GAN-based model to generate realistic fault scenarios in power grids, providing a means to train predictive models when real fault data is scarce or incomplete. This study distinguishes itself by synergistically combining generative adversarial models and graph-based neural architectures to predict faults within interconnected energy systems. The generative component synthesizes plausible fault events, while the graph learning model identifies intricate dependencies among electricity, gas, and thermal networks, thereby enhancing the accuracy of fault propagation forecasting.

Another significant area of research in fault management involves optimization techniques for fault recovery, particularly under conditions of uncertainty. Robust optimization has long been used to handle uncertainty in operational conditions, and it has been applied to both energy generation and distribution systems. However, traditional robust optimization models often rely on fixed uncertainty sets and may not account for the dynamic nature of fault propagation[22,23]. These limitations have led researchers to explore DRO, which allows for uncertainty sets that are distributionally flexible and adaptable to changing conditions. Reference[24] provided a comprehensive framework for DRO, demonstrating its effectiveness in optimizing decision-making under uncertain conditions. In the context of energy networks, DRO has been used to optimize power generation and distribution in the presence of uncertain demand and supply conditions. For example[25], applied DRO to optimize the scheduling of energy resources in microgrids, accounting for uncertainties in renewable energy generation. Similarly[26], used DRO to optimize recovery strategies in power systems following natural disasters, showing that DRO outperformed traditional robust optimization by accounting for more realistic uncertainty distributions[27].

While significant progress has been made in modeling fault propagation, predicting faults with AI, and optimizing recovery strategies under uncertainty, this paper provides several key advancements that address existing gaps in the literature. First, unlike previous work that typically focuses on fault management in isolated networks, this paper presents a holistic framework that integrates electric, gas, and heating networks[28]. This novel approach accounts for the interdependencies between these systems, allowing for a more comprehensive understanding of how faults propagate across multiple networks. Second, the integration of GANs and GNNs represents a novel approach to fault prediction. GANs are used to generate realistic fault propagation scenarios, which serve as training data for the GNN[29]. The GNN then models the dynamic relationships between nodes in the energy networks, predicting how faults will propagate over time and space. This combination of GANs and GNNs is unique in the field and offers significant improvements in prediction accuracy[30]. Third, the use of DRO in this paper extends beyond traditional applications by optimizing fault recovery strategies across interconnected energy networks. DRO ensures that recovery strategies are robust to the uncertainties inherent in fault propagation, providing a more resilient solution to fault management. In conclusion, this study advances the expanding field of AI-based fault management by introducing an integrated framework that simultaneously

models fault propagation and formulates resilient recovery strategies within a cohesive system architecture. The integration of AI techniques and robust optimization methodologies sets this research apart from existing approaches and provides a scalable, practical solution for managing faults in real-world energy systems[31].

Mathematical modeling

In this section, we present a comprehensive mathematical formulation for optimizing the operation of an integrated energy system comprising electricity, gas, and heating networks. The main goal is to achieve a reduction in the overall operational expenditure, encompassing costs associated with energy generation, system maintenance, and efficiency losses.

$$C_{\text{impact}}(t) = \sum_{i=1}^{N_{\text{elec}}} \alpha_i \cdot \xi_i(t) + \sum_{j=1}^{N_{\text{gas}}} \beta_j \cdot \psi_j(t) + \sum_{k=1}^{N_{\text{heat}}} \gamma_k \cdot \zeta_k(t) + \lambda_{\text{elec}} \cdot \Delta_{\text{elec}}(t) + \lambda_{\text{gas}} \cdot \Delta_{\text{gas}}(t) + \lambda_{\text{heat}} \cdot \Delta_{\text{heat}}(t) \quad (1)$$

Equation (1) shows the cumulative fault impact cost $C_{\text{impact}}(t)$ at time t, calculated by summing the weighted fault severities of the electrical network $\xi_i(t)$ with weights α_i, the gas network $\psi_j(t)$ with weights β_j, and the heating network $\zeta_k(t)$ with weights γ_k, as well as the weighted service loss terms $\Delta_{\text{elec}}(t)$, $\Delta_{\text{gas}}(t)$, and $\Delta_{\text{heat}}(t)$ with respective penalty coefficients λ_{elec}, λ_{gas}, and λ_{heat}.

The expression for system-level fault impact cost quantifies the combined severity of faults occurring across electric, gas, and heating networks. In intuitive terms, it measures how much each fault degrades the system, weighted by its location and operational importance. Each network contributes a time-varying disturbance, which is scaled by a priority factor to reflect its criticality. Additionally, penalty terms are included to represent service interruptions, ensuring that both direct damage and indirect loss of functionality are considered in the impact assessment.

$$P_{uv}(t) = \sigma\left(\kappa_1 \cdot h_u(t) + \kappa_2 \cdot h_v(t) + \kappa_3 \cdot \eta_{uv} + \kappa_4 \cdot d(u,v) + \kappa_5 \cdot \gamma_u(t) + \kappa_6 \cdot \gamma_v(t)\right) \quad (2)$$

Equation (2) represents the probability $P_{uv}(t)$ of fault propagation from node u to node v, computed through a sigmoid function $\sigma(\cdot)$ applied to a weighted sum of the local features $h_u(t)$ and $h_v(t)$ with weights κ_1 and κ_2, the edge attribute η_{uv} weighted by κ_3, the physical distance $d(u, v)$ weighted by κ_4, and the vulnerability features $\gamma_u(t)$ and $\gamma_v(t)$ weighted by κ_5 and κ_6.

The fault propagation probability between two components is modeled as a function of their internal condition, interconnection characteristics, and physical proximity. In essence, the model assesses whether a vulnerable node is likely to be affected by its neighboring node, factoring in how closely they are linked and how susceptible each is at a given time. The strength of the connection, the operational stress level, and the network distance all influence the likelihood of propagation.

$$P_i(t) = \sum_{j=1}^{N_{\text{elec}}} V_i(t)V_j(t)\left(G_{ij}\cos(\theta_i(t) - \theta_j(t)) + B_{ij}\sin(\theta_i(t) - \theta_j(t))\right) + \epsilon_i(t) + \delta_i(t) + \eta_i(t) + \rho_i(t) + \chi_i(t) \quad (3)$$

Equation (3) describes the power balance at electrical node i, where $P_i(t)$ is the sum of power exchanges with neighboring nodes j considering voltage magnitudes $V_i(t)$, $V_j(t)$, conductance G_{ij}, susceptance B_{ij}, and phase angle differences $\theta_i(t) - \theta_j(t)$, with additional disturbance terms $\epsilon_i(t)$, $\delta_i(t)$, $\eta_i(t)$, and recovery adjustment terms $\rho_i(t)$ and $\chi_i(t)$.

$$\min_{r(t)} \quad \sum_{i=1}^{N_{\text{elec}}} \alpha_i \left(r_i(t) - r_i^*(t)\right)^2 + \sum_{j=1}^{N_{\text{gas}}} \beta_j \left(r_j(t) - r_j^*(t)\right)^2 + \sum_{k=1}^{N_{\text{heat}}} \gamma_k \left(r_k(t) - r_k^*(t)\right)^2 + \phi_1\|r(t)\|_1 + \phi_2\|r(t)\|_2^2 + \phi_3 \cdot \text{Var}(r(t)) \quad (4)$$

Equation (4) formulates a recovery resource optimization problem aiming to minimize the quadratic deviation between the allocated recovery amounts $r_i(t), r_j(t), r_k(t)$ and their targets $r_i^*(t), r_j^*(t), r_k^*(t)$ for electrical, gas, and heat nodes respectively, while also introducing ℓ_1-norm regularization $\|r(t)\|_1$ weighted by ϕ_1, ℓ_2-norm regularization $\|r(t)\|_2^2$ weighted by ϕ_2, and variance regularization $\text{Var}(r(t))$ weighted by ϕ_3. The recovery allocation model is designed to distribute available resources–such as repair crews or control adjustments–in a way that minimizes the deviation from ideal restoration targets. It not only penalizes under- or over-allocation but also encourages smooth, coordinated recovery by introducing penalties on abrupt changes and uneven distribution. This reflects the practical need for stable and cost-efficient recovery strategies during multi-network disturbances.

$$\min_{u(t)} \max_{\mathbb{Q}\in\mathcal{U}} \mathbb{E}_{\mathbb{Q}}\left[\sum_{t=1}^{T}\left(C_{\text{impact}}(t) + C_{\text{rec}}(t) + \eta_1\|u(t)\|_2^2 + \eta_2\|\nabla u(t)\|_2^2 + \eta_3\Delta_{\text{resilience}}(t)\right)\right] \quad (5)$$

Equation (5) shows a distributionally robust optimization model where the controller seeks $u(t)$ to minimize the worst-case expected cost over all distributions $\mathbb{Q}$ in the uncertainty set $\mathcal{U}$, considering the fault impact cost $C_{\text{impact}}(t)$, recovery cost $C_{\text{rec}}(t)$, control energy $\|u(t)\|_2^2$ weighted by η_1, control variation $\|\nabla u(t)\|_2^2$ weighted by η_2, and resilience degradation $\Delta_{\text{resilience}}(t)$ weighted by η_3.

$$\min_{\mathrm{u}(t)} \sum_{n\in\{\mathrm{elec,gas,heat}\}} \left(\sum_{i=1}^{N_n} (\chi_{n,i}(t)\cdot\phi_{n,i}(t) + \gamma_{n,i}(t)\cdot\sigma_{n,i}(t) + \beta_{n,i}(t)\cdot\Delta_{n,i}(t)) \right) + \eta\cdot\|\nabla \mathrm{u}(t)\|_2^2 + \xi\cdot \mathrm{Var}(\mathrm{u}(t)) + \rho\cdot\|\mathrm{u}(t)\|_1 \quad (6)$$

Equation (6) describes the minimization of total network loss across electricity, gas, and heat networks, where each node contributes weighted terms $\chi_{n,i}(t)\phi_{n,i}(t)$, $\gamma_{n,i}(t)\sigma_{n,i}(t)$, and $\beta_{n,i}(t)\Delta_{n,i}(t)$, with additional penalties on control variation $\|\nabla \mathrm{u}(t)\|_2^2$ weighted by η, control variance $\mathrm{Var}(\mathrm{u}(t))$ weighted by ξ, and control sparsity $\|\mathrm{u}(t)\|_1$ weighted by ρ.

$$h_v^{(L)} = \sigma\left(W^{(L)} \left(\sum_{l=1}^{L-1} \alpha^{(l)} h_v^{(l)} + \sum_{u\in\mathcal{N}(v)} \beta^{(l)} h_u^{(l)} + \gamma^{(l)} \mathrm{e}_{uv} + \delta^{(l)} d(u,v) + \epsilon^{(l)} \mathrm{type}(v) \right) \right) \quad (7)$$

Equation (7) shows the update rule for node embedding $h_v^{(L)}$ at layer L in a graph neural network, combining previous layer features $h_v^{(l)}$ and $h_u^{(l)}$ from neighboring nodes $u \in \mathcal{N}(v)$, edge features e_{uv}, physical distances $d(u, v)$, and node type encoding $\mathrm{type}(v)$, with respective weights $\alpha^{(l)}$, $\beta^{(l)}$, $\gamma^{(l)}$, $\delta^{(l)}$, and $\epsilon^{(l)}$, followed by transformation matrix $W^{(L)}$ and activation $\sigma(\cdot)$.

$$\mathcal{R}_{\mathrm{fault}}(t) = \sum_{s=1}^{S} p_s \left(\sum_{n} \sum_{i=1}^{N_n} \lambda_{n,i} \cdot \left(\xi_{n,i}^{(s)}(t) + \delta_{n,i}^{(s)}(t) + \psi_{n,i}^{(s)}(t) + \zeta_{n,i}^{(s)}(t) + \kappa_{n,i}^{(s)}(t) + \epsilon_{n,i}^{(s)}(t) \right) \right) \quad (8)$$

Equation (8) defines the fault risk functional $\mathcal{R}_{\mathrm{fault}}(t)$ under a scenario-based framework, summing over S different fault scenarios each weighted by probability p_s, and for each scenario aggregating fault disturbances $\xi_{n,i}^{(s)}(t)$, $\delta_{n,i}^{(s)}(t)$, $\psi_{n,i}^{(s)}(t)$, $\zeta_{n,i}^{(s)}(t)$, $\kappa_{n,i}^{(s)}(t)$, and $\epsilon_{n,i}^{(s)}(t)$ weighted by node-specific sensitivity $\lambda_{n,i}$.

$$\mathrm{u}(t+1) = \mathrm{u}(t) + \alpha\cdot\nabla_{\mathrm{u}} C_{\mathrm{total}}(t) + \beta\cdot\Delta_{\mathrm{fault}}(t) + \gamma\cdot\Delta_{\mathrm{topo}}(t) + \delta\cdot\Delta_{\mathrm{load}}(t) + \epsilon\cdot\zeta(t) + \phi\cdot\nabla_{\mathrm{u}}^2 C_{\mathrm{total}}(t) \quad (9)$$

Equation (9) represents the adaptive update of the control action $\mathrm{u}(t+1)$ by adding the scaled gradients $\nabla_{\mathrm{u}} C_{\mathrm{total}}(t)$ with coefficient α, the scaled fault variation $\Delta_{\mathrm{fault}}(t)$ with coefficient β, the scaled topology change $\Delta_{\mathrm{topo}}(t)$ with coefficient γ, the scaled load fluctuation $\Delta_{\mathrm{load}}(t)$ with coefficient δ, the scaled noise $\zeta(t)$ with coefficient ϵ, and the scaled second-order gradient correction $\nabla_{\mathrm{u}}^2 C_{\mathrm{total}}(t)$ with coefficient ϕ.

$$\mathcal{F}_{\mathrm{risk}}(t) = \omega_1\cdot\mathrm{E}[\Delta s(t)] + \omega_2\cdot\mathrm{Var}(\Delta s(t)) + \omega_3\cdot\sum_{v\in V}\left(\lambda_v\cdot\mathbb{I}\{s_v(t) < s_v^{\min}\}\right) + \omega_4\cdot\|\mathrm{x}(t) - \mathrm{x}(t-1)\|_2^2 + \omega_5\cdot\|\nabla \mathrm{x}(t)\|_2^2 + \omega_6\cdot\mathrm{Entropy}(s(t)) \quad (10)$$

Equation (10) shows the system-level fault risk metric $\mathcal{F}_{\mathrm{risk}}(t)$, which aggregates the expected service degradation $\mathrm{E}[\Delta s(t)]$ weighted by ω_1, the service variance $\mathrm{Var}[\Delta s(t)]$ weighted by ω_2, the cumulative penalty for nodes violating minimum service levels $\mathbb{I}\{s_v(t) < s_v^{\min}\}$ weighted by ω_3, the control smoothness penalties $\|\mathrm{x}(t) - \mathrm{x}(t-1)\|_2^2$ and $\|\nabla \mathrm{x}(t)\|_2^2$ weighted by ω_4 and ω_5, and the system entropy $\mathrm{Entropy}(s(t))$ weighted by ω_6.

$$\sum_{i=1}^{N_{\mathrm{elec}}} P_i^{\mathrm{gen}}(t) + \sum_{j=1}^{N_{\mathrm{gas}}} p_j^{\mathrm{gen}}(t) = \sum_{m=1}^{N_{\mathrm{load}}} d_m(t) + \Delta_{\mathrm{fault}}(t) \quad (11)$$

Equation (11) shows the energy balance in the multi-energy system, where the sum of electrical generation $P_i^{\mathrm{gen}}(t)$ and gas generation $p_j^{\mathrm{gen}}(t)$ must equal the total demand $\sum_{m=1}^{N_{\mathrm{load}}} d_m(t)$ plus the fault-induced loss $\Delta_{\mathrm{fault}}(t)$ at time t.

$$\mathcal{J}(t) = \|u(t) - u^*(t)\|_2^2 + \alpha_1\|\dot{u}(t)\|_2^2 + \alpha_2\|x(t) - x^{\mathrm{target}}(t)\|_2^2 + \alpha_3\mathrm{Risk}(t) \quad (12)$$

Equation (12) represents the recovery trajectory cost $\mathcal{J}(t)$, which is composed of the deviation between the actual control $u(t)$ and the target control $u^*(t)$, penalized by the quadratic norm, the squared control variation $\|\dot{u}(t)\|_2^2$ weighted by α_1, the state deviation $\|x(t) - x^{\mathrm{target}}(t)\|_2^2$ weighted by α_2, and the associated risk $\mathrm{Risk}(t)$ weighted by α_3.

$$\delta_v(t) = \sigma\left(W h_v^{(L)} + \sum_{u\in\mathcal{N}(v)} \beta_{uv} h_u^{(L)} + e_{uv} + \eta_v(t) \right) \quad (13)$$

Equation (13) describes the fault disturbance generation $\delta_v(t)$ at node v using a GNN structure, where the node's own embedding $h_v^{(L)}$, aggregated neighbor embeddings $\sum_{u\in\mathcal{N}(v)} \beta_{uv} h_u^{(L)}$, edge feature e_{uv}, and local noise $\eta_v(t)$ are linearly combined with weight matrix W and passed through a sigmoid activation function $\sigma(\cdot)$.

$$\min_{u(t)} \sum_{t} \left(C_{\text{op}}(t) + C_{\text{fault}}(t) + \lambda_1 \|u(t) - u^{\text{nominal}}(t)\|_2^2 + \lambda_2 \|\nabla u(t)\|_2^2 \right) \tag{14}$$

Equation (14) shows the fault-resilient dispatch optimization, where the objective is to minimize the sum of the normal operation cost $C_{\text{op}}(t)$, the fault cost $C_{\text{fault}}(t)$, the control deviation penalty $\|u(t) - u^{\text{nominal}}(t)\|_2^2$ weighted by λ_1, and the control variation penalty $\|\nabla u(t)\|_2^2$ weighted by λ_2 over the dispatch horizon.

$$x_v(t+1) = f(x_v(t), u_v(t)) + g\left(\sum_{u \in \mathcal{N}(v)} W_{uv}(t) x_u(t) + B_{uv}(t) \delta_u(t) \right) \tag{15}$$

Equation (15) describes the time-coupled dynamics of node v, where the next state $x_v(t+1)$ depends on the current state $x_v(t)$ and control input $u_v(t)$ through a function $f(\cdot)$, and also aggregates the neighboring nodes' states $x_u(t)$ and their fault disturbances $\delta_u(t)$ through weighted functions $g(\cdot)$ involving weight matrices $W_{uv}(t)$ and $B_{uv}(t)$.

$$\sum_{i=1}^{N_{\text{elec}}} P_i(t) + \sum_{j=1}^{N_{\text{gas}}} p_j(t) + \sum_{k=1}^{N_{\text{heat}}} h_k(t) = D_{\text{total}}(t) + \Delta_{\text{loss}}(t) \tag{16}$$

Equation (16) shows the simplified energy balance equation in a multi-energy system, where the sum of electrical $P_i(t)$, gas $p_j(t)$, and heat $h_k(t)$ outputs should match the total load demand $D_{\text{total}}(t)$ plus the system loss term $\Delta_{\text{loss}}(t)$ at time t.

$$C_{\text{rec}}(t) = \sum_{v \in V} \left(\alpha_v \|u_v(t) - u_v^*(t)\|^2 + \beta_v \|\dot{u}_v(t)\|^2 + \gamma_v \text{Deviation}(x_v(t)) \right) \tag{17}$$

Equation (17) represents the recovery cost $C_{\text{rec}}(t)$ as the aggregation over all nodes v of three terms: the quadratic control deviation $\|u_v(t) - u_v^*(t)\|^2$ weighted by α_v, the control variation $\|\dot{u}_v(t)\|^2$ weighted by β_v, and the deviation of the state $x_v(t)$ from a desired condition captured by $\text{Deviation}(x_v(t))$ weighted by γ_v.

$$x_v(t+1) = A_v x_v(t) + B_v u_v(t) + D_v \delta_v(t) + \epsilon_v(t) \tag{18}$$

Equation (18) describes the state transition for node v, where the next state $x_v(t+1)$ is modeled linearly based on the current state $x_v(t)$ multiplied by the system matrix A_v, the control input $u_v(t)$ multiplied by control matrix B_v, the fault disturbance $\delta_v(t)$ through the disturbance matrix D_v, and the additive noise term $\epsilon_v(t)$.

$$s_v(t) = \text{MLP}\left(h_v^{(L)} + \sum_{u \in \mathcal{N}(v)} e_{uv} + d(u, v) \right) \tag{19}$$

Equation (19) shows the estimation of the fault score $s_v(t)$ for node v, where a multi-layer perceptron (MLP) processes the combination of the node's embedding $h_v^{(L)}$, the sum of edge features $\sum_{u \in \mathcal{N}(v)} e_{uv}$, and the physical distance $d(u, v)$ between node v and its neighbors.

$$\min_{u(t)} \sum_{t=1}^{T} \left(C_{\text{op}}(t) + C_{\text{rec}}(t) + \lambda \text{Risk}(t) + \rho \|u(t)\|_1 \right) \tag{20}$$

Equation (20) defines the system-wide recovery objective, where the controller minimizes the sum of the normal operation cost $C_{\text{op}}(t)$, the recovery cost $C_{\text{rec}}(t)$, the risk term $\text{Risk}(t)$ weighted by λ, and the ℓ_1-norm sparsity term $\|u(t)\|_1$ weighted by ρ over the entire recovery horizon.

$$\mathcal{L}(t) = C_{\text{op}}(t) + \gamma_1 \text{Risk}_{\text{fault}}(t) + \gamma_2 \|\nabla u(t)\|_2^2 + \gamma_3 \|u(t)\|_1 \tag{21}$$

Equation (21) shows the dynamic risk-adjusted dispatch objective $\mathcal{L}(t)$ at each time step, which includes the operational cost $C_{\text{op}}(t)$, the fault risk $\text{Risk}_{\text{fault}}(t)$ weighted by γ_1, the control variation penalty $\|\nabla u(t)\|_2^2$ weighted by γ_2, and the sparsity-promoting ℓ_1-norm term $\|u(t)\|_1$ weighted by γ_3.

$$\mathcal{L}_{\text{fault}}(t) = \sum_{v \in V} \left(\phi_v(t) + \psi_v(t) + \eta_v(t) + \xi_v(t) \right) \tag{22}$$

Equation (22) represents the aggregated fault propagation loss $\mathcal{L}_{\text{fault}}(t)$, where at each node v, four types of disturbance factors $\phi_v(t)$, $\psi_v(t)$, $\eta_v(t)$, and $\xi_v(t)$ are summed to reflect the combined local fault impacts.

$$s_v(t) \geq s_v^{\min} - \alpha \delta_v(t) - \beta \epsilon_v(t) - \gamma \eta_v(t) \tag{23}$$

Equation (23) describes the service level constraint under faults, where the service level $s_v(t)$ of node v must remain above the minimum threshold $s_v^{\min}$, offset by the fault disturbances $\delta_v(t)$, $\epsilon_v(t)$, and $\eta_v(t)$ weighted by coefficients α, β, and γ respectively.

$$\delta_v^{(s)}(t) = \mu_v(t) + \sigma_v(t) \cdot z^{(s)}(t), \quad z^{(s)}(t) \sim \mathcal{N}(0,1) \tag{24}$$

Equation (24) shows the sampling process for stochastic fault scenarios, where the fault disturbance $\delta_v^{(s)}(t)$ is generated by adding Gaussian noise $z^{(s)}(t) \sim \mathcal{N}(0,1)$ scaled by the standard deviation $\sigma_v(t)$ to the mean fault level $\mu_v(t)$.

$$\min_{u(t)} \max_{\mathbb{Q}\in\mathcal{U}} \mathbb{E}_{\mathbb{Q}} \left[\sum_{t=1}^{T} C_{\text{total}}(t) \right] \tag{25}$$

Equation (25) represents the distributionally robust recovery formulation, where the control policy $u(t)$ minimizes the worst-case expected total cost $\sum_{t=1}^{T} C_{\text{total}}(t)$ across all possible probability distributions $\mathbb{Q}$ contained in the uncertainty set $\mathcal{U}$.

This chapter has established a comprehensive mathematical framework for modeling fault propagation and recovery optimization across interconnected electricity, gas, and heating networks. The proposed approach integrates generative adversarial models to synthesize realistic fault scenarios, graph-based neural networks to capture spatiotemporal fault dynamics, and a distributionally robust optimization structure to formulate resilient recovery strategies under uncertainty. The formulation systematically addresses system state representation, multi-network coupling, fault-induced disturbances, operational cost decomposition, and service-level constraints, ensuring a holistic description of the system behaviors under adverse conditions. By incorporating both stochastic fault generation and time-coupled propagation effects, the model provides a flexible and scalable foundation for subsequent learning-based optimization procedures. The developed modeling framework not only enhances the fidelity of fault impact estimation but also enables adaptive and risk-averse recovery planning. Building upon this foundation, the next chapter will detail the algorithmic methodology and solution techniques applied to solve the formulated optimization problem efficiently. While the proposed integration of generative adversarial networks, graph neural networks, and distributionally robust optimization offers strong theoretical performance and predictive capabilities, its practical implementation in real-world energy systems poses several challenges. One key issue is the requirement for high-quality, time-synchronized multi-network data–including fault logs, system states, and inter-network coupling features–which are often incomplete or inconsistently recorded in real infrastructures. This limits the direct applicability of end-to-end training unless data augmentation or transfer learning techniques are employed.

Additionally, the computational resources needed for model training and online deployment are non-trivial. The generative model and graph learning modules must be trained on large-scale historical scenarios, and the robust optimization layer involves iterative solvers that may not meet real-time constraints in high-frequency operational settings. These factors highlight the importance of developing lightweight approximations, edge-deployable variants, or modular implementations to bridge the gap between advanced AI-based modeling and operational feasibility.

LSTM-enhanced framework for temporal state inference

This section introduces the application of Long Short-Term Memory (LSTM) networks for dynamic state estimation within the integrated energy system. LSTM, a type of recurrent neural network, is particularly suited for modeling temporal dependencies in time-series data such as energy consumption and generation patterns. By utilizing past information through memory cells, LSTMs can accurately predict future system states, aiding in decision-making for real-time control and optimization. We describe the mathematical foundations of the LSTM model, including the equations governing hidden state updates, input and forget gates, and output computation. The integration of LSTM into the system's predictive framework enables a more adaptive and responsive energy management system capable of handling complex, nonlinear behaviors across electricity, gas, and heat networks.

$$i_t = \sigma\left(W_i x_t + U_i h_{t-1} + b_i\right) \tag{26}$$

Equation (26) shows the computation of the input gate i_t in the LSTM unit, where the current input x_t and the previous hidden state h_{t-1} are linearly transformed using weights W_i and U_i, added with a bias term b_i, and passed through a sigmoid activation function $\sigma(\cdot)$ to determine the extent of new information flow into the cell state.

$$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \tag{27}$$

Equation (27) shows the computation of the forget gate f_t, where the same structure as the input gate is used: the input x_t and previous hidden state h_{t-1} are transformed with weights W_f and U_f, and a bias b_f, then processed through a sigmoid activation $\sigma(\cdot)$ to control the memory retention from the previous cell state c_{t-1}.

$$\tilde{c}_t = \tanh\left(W_c x_t + U_c h_{t-1} + b_c\right) \tag{28}$$

Equation (28) shows the generation of the candidate cell state $\tilde{c}_t$, where x_t and h_{t-1} are transformed through weights W_c and U_c, summed with bias b_c, and activated by a hyperbolic tangent function $\tanh(\cdot)$ to form the new memory content proposal.

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \tag{29}$$

Equation (29) shows the update of the actual cell state c_t, where the previous memory c_{t-1} is partially retained by the forget gate f_t, and the candidate memory $\tilde{c}_t$ is partially incorporated by the input gate i_t. The operations are element-wise multiplications $\odot$.

$$h_t = o_t \odot \tanh(c_t), \quad o_t = \sigma\left(W_o x_t + U_o h_{t-1} + b_o\right) \tag{30}$$

Equation (30) shows the computation of the hidden state h_t, where the updated cell state c_t is first activated through $\tanh(\cdot)$, then modulated by the output gate o_t. The output gate itself is computed based on the current input x_t, the previous hidden state h_{t-1}, the weight matrices W_o, U_o, and the bias b_o through a sigmoid activation.

$$\mathrm{z}_t = \mathrm{Concat}\left(x_t, u_t, \delta_t\right) \tag{31}$$

Equation (31) shows the construction of the input vector z_t to the LSTM, which concatenates the current system observation x_t, the applied control input u_t, and the disturbance or fault input δ_t into a single feature vector for sequential modeling.

$$\hat{x}_t = \mathcal{F}_{\mathrm{LSTM}}(\mathrm{z}_{1:t}) \tag{32}$$

Equation (32) shows the estimation of the system state $\hat{x}_t$ at time t, where the LSTM-based model $\mathcal{F}_{\mathrm{LSTM}}$ processes the sequence of inputs $\mathrm{z}_{1:t}$ from time 1 up to t to infer the latent system dynamics and output the predicted state.

$$\mathcal{L}_{\mathrm{pred}} = \frac{1}{T}\sum_{t=1}^{T}\left\|\hat{x}_t - x_t^{\mathrm{true}}\right\|_2^2 \tag{33}$$

Equation (33) shows the definition of the prediction loss $\mathcal{L}_{\mathrm{pred}}$, computed as the mean squared error (MSE) between the estimated states $\hat{x}_t$ and the true states x_t^{true} over the entire time horizon T.

$$\mathcal{R}_{\mathrm{temp}} = \frac{1}{T-1}\sum_{t=2}^{T}\|\hat{x}_t - \hat{x}_{t-1}\|_2^2 \tag{34}$$

Equation (34) shows the temporal consistency regularization term $\mathcal{R}_{\mathrm{temp}}$, which encourages the predicted states $\hat{x}_t$ to vary smoothly over time by penalizing the squared differences between consecutive predictions $\hat{x}_t$ and $\hat{x}_{t-1}$.

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{pred}} + \lambda_{\mathrm{temp}}\mathcal{R}_{\mathrm{temp}} \tag{35}$$

Equation (35) shows the total training loss $\mathcal{L}_{\mathrm{total}}$, formed by combining the prediction loss $\mathcal{L}_{\mathrm{pred}}$ and the temporal regularization term $\mathcal{R}_{\mathrm{temp}}$, weighted by a balancing coefficient λ_{temp} to trade off accuracy and temporal smoothness.

$$h_t = \mathcal{H}(h_{t-1}, \mathrm{z}_t) \tag{36}$$

Equation (36) shows the hidden state propagation rule in the LSTM, where the current hidden state h_t is obtained by a function $\mathcal{H}(\cdot)$ applied to the previous hidden state h_{t-1} and the current input vector z_t, encoding the sequential dependency.

$$c_t = \mathcal{C}(c_{t-1}, i_t, f_t, \tilde{c}_t) \tag{37}$$

Equation (37) shows the update of the cell state c_t through a function $\mathcal{C}(\cdot)$, which depends on the previous cell state c_{t-1}, the input gate i_t, the forget gate f_t, and the candidate cell update $\tilde{c}_t$, following the standard LSTM update rule.

$$\hat{x}_t = W_o h_t + b_o \tag{38}$$

Equation (38) shows the mapping from the hidden state h_t to the predicted system state $\hat{x}_t$, using a simple affine transformation with output weight matrix W_o and bias vector b_o to project the hidden features to the original state space.

$$\hat{x}_{t+\tau} = \mathcal{F}_{\mathrm{LSTM}}(\mathrm{z}_t, \ldots, \mathrm{z}_{t+\tau-1}) \tag{39}$$

Equation (39) shows the extension of the LSTM to multi-step forecasting, where the future state $\hat{x}_{t+\tau}$ at step τ is predicted based on the input sequence from time t to $t+\tau-1$, processed through the LSTM encoder $\mathcal{F}_{\mathrm{LSTM}}$.

$$\hat{x}_{t+\tau} = \mathcal{F}\left(\hat{x}_{t+\tau-1}, u_{t+\tau-1}, \delta_{t+\tau-1}\right) \tag{40}$$

Equation (40) shows the recursive prediction strategy for long-term forecasting, where the next predicted state $\hat{x}_{t+\tau}$ is inferred by feeding the previous predicted state $\hat{x}_{t+\tau-1}$ along with the associated control input $u_{t+\tau-1}$ and disturbance $\delta_{t+\tau-1}$ into a prediction function $\mathcal{F}(\cdot)$.

$$\mathcal{L}_{\mathrm{multi}} = \frac{1}{T}\sum_{\tau=1}^{T}\left\|\hat{x}_{t+\tau} - x_{t+\tau}^{\mathrm{true}}\right\|_2^2 \tag{41}$$

Equation (41) shows the multi-step prediction loss $\mathcal{L}_{\mathrm{multi}}$, calculated as the average squared error between multi-step predicted states $\hat{x}_{t+\tau}$ and their ground truth counterparts $x_{t+\tau}^{\mathrm{true}}$ across the full horizon T.

$$\mathcal{L}_{\mathrm{weighted}} = \sum_{\tau=1}^{T}\omega(\tau)\left\|\hat{x}_{t+\tau} - x_{t+\tau}^{\mathrm{true}}\right\|_2^2 \tag{42}$$

Equation (42) shows the adaptive weighted prediction loss $\mathcal{L}_{\mathrm{weighted}}$, where each prediction error at horizon τ is weighted by a function $\omega(\tau)$, allowing the model to prioritize errors at shorter horizons or adjust the importance dynamically.

$$\mathrm{z}_{t+\tau} = (1-\epsilon)\cdot \text{real input} + \epsilon \cdot \text{predicted input} \tag{43}$$

Equation (43) shows the scheduled sampling mechanism, where the input at time $t+\tau$ is a convex combination of the real ground truth input and the model-predicted input, controlled by a probability ϵ, in order to gradually shift the model from teacher-forcing to free-running during training.

$$\tilde{\mathrm{z}}_t = \mathrm{z}_t + \eta_t, \quad \eta_t \sim \mathcal{N}(0, \sigma^2 I) \tag{44}$$

Equation (44) shows the noise injection strategy for training robustness, where a Gaussian noise η_t sampled from a normal distribution $\mathcal{N}(0,\sigma^2 I)$ is added to the original input z_t to obtain the perturbed input $\tilde{\mathrm{z}}_t$, enhancing generalization under disturbances.

$$\mathrm{Conf}_t = \sigma\left(W_c h_t + b_c\right) \tag{45}$$

Equation (45) shows the dynamic confidence estimation Conf_t for the predicted output at time t, obtained by applying a sigmoid function $\sigma(\cdot)$ to a linear transformation of the hidden state h_t with weight W_c and bias b_c, representing the model's internal certainty about its prediction.

Results

In this case study, the data used for simulating fault propagation across the interconnected electric, gas, and heating networks is synthesized and assumed based on reasonable estimates and existing standards in the energy industry. The electric network is modeled using the IEEE 123-bus system, which represents a typical low-voltage distribution system with 123 nodes. Each node represents a distribution point or substation, and the system includes 154 branches connecting the nodes. The assumed nominal voltage for the system is 13.8 kV, with each node being assigned a power demand ranging from 50 to 500 kW, reflecting a mix of residential, commercial, and industrial loads. The total system demand is synthesized to reach approximately 40 MW during peak hours, while the generation capacity is distributed across five synthetic generators, each with a maximum output ranging from 5 to 15 MW. The gas network is based on the Belgian gas transmission system, which includes 20 major nodes and 30 pipelines. The synthesized data assumes that the maximum gas flow per pipeline ranges between 50 and 150 MWh/day. Each node is assigned a gas demand of 10–60 MWh/day, depending on its assumed industrial or residential status. The gas pressure at each node is constrained to remain between 30 and 70 bar, ensuring that the gas supply remains stable during both normal and fault conditions. The heating network follows a standard district heating system, with 15 nodes and a synthesized total heat demand of 100 MW, where each node has an assumed demand ranging between 2 MW and 10 MW. Heat is delivered through a series of heat exchangers, with maximum flow rates set at 10 tons per hour[4,27–29]. The computational experiments for the case study are performed in a high-performance computing (HPC) environment equipped with 32 Intel Xeon processors (3.6 GHz) and 256 GB of RAM, ensuring sufficient processing power to handle the complex interactions between the networks. The GAN and GNN models are developed using Python, leveraging the TensorFlow and PyTorch libraries for machine learning tasks. The GAN requires training on a dataset of approximately 10,000 synthetic fault scenarios, with each scenario representing fault propagation over a 24-h period at 15-min intervals. Training the GAN and GNN models is computationally intensive, requiring an estimated 8 h for full training, with batch sizes of 128 and an initial learning rate of 0.001. The DRO model, used for optimizing the recovery strategies, is implemented in MATLAB, utilizing the CVX optimization toolbox. The DRO model incorporates uncertainty in the form of 500 different fault propagation paths, each representing potential variations in the fault spread between the networks. The computation of each optimization scenario

takes approximately 10 min, with the total time for running all scenarios estimated at 85 h on the HPC system. All simulations are conducted under Linux (Ubuntu 20.04) for stability and performance, with results exported into CSV files for further analysis and visualization using MATLAB and Python.

To simulate fault propagation, the system assumes a variety of fault scenarios, including single-node failures, cascading multi-node faults, and cross-network interactions where an initial fault in the electric grid leads to failures in the gas and heating networks. Fault scenarios are injected at 10 randomly selected nodes in each network, with a fault duration ranging between 5 and 60 min. The interdependent nature of the networks is highlighted by adjusting the fault propagation speed and severity, with propagation times between networks set to vary between 1 and 10 min. The case study is designed to capture the cascading effects of faults under different load, demand, and recovery conditions, allowing the GAN and GNN models to simulate realistic and complex fault dynamics across the interconnected networks.

Figure 2 shows the synthesized power demand data for each node in the 123-bus system, illustrating the spread of power requirements across residential, commercial, and industrial nodes. The power demand values are grouped into three distinct categories: residential nodes, which have demands ranging between 50 and 100 kW; commercial nodes, with demands between 100 and 300 kW; and industrial nodes, which have the highest demand, ranging between 300 and 500 kW. From the histogram, we can observe that the majority of nodes have power demands concentrated between 50 and 150 kW, with a peak near 75 kW, corresponding to residential consumption patterns. The total synthesized demand across all nodes is approximately 40 MW, reflecting a mix of diverse load profiles within the system. The distribution of power demands is critical to understanding how load imbalances may propagate through the network during fault scenarios. Nodes with larger power demands, such as those in the industrial category with demand nearing 500 kW, can exert significant strain on the system if they are suddenly disconnected or disrupted. These industrial nodes, located toward the right end of the histogram, represent roughly 25% of the total system demand. In contrast, the residential nodes, which cluster toward the left with demands closer to 50 kW, are individually smaller consumers but collectively represent a significant portion of the network's demand. The fact that residential nodes constitute around 40% of the total node count suggests that a distributed fault event affecting many smaller nodes could still cause significant disruptions, albeit in a different manner compared to industrial node failures. Additionally, the commercial nodes, with demand ranging between 100 and 300 kW, form an important middle category in the system's demand profile. The histogram shows that commercial nodes represent about 33% of the total system nodes, with their demands more evenly spread across the 100–300 kW range. These nodes typically serve businesses, office complexes, and small manufacturing facilities. While each node's demand is lower than industrial nodes, they form a significant part of the grid's load. This even distribution helps to stabilize the system, as commercial nodes do not contribute to peak loads as much as industrial nodes, but they also provide a more constant load compared to the variability often seen in residential demand. Thus, understanding the distribution of power

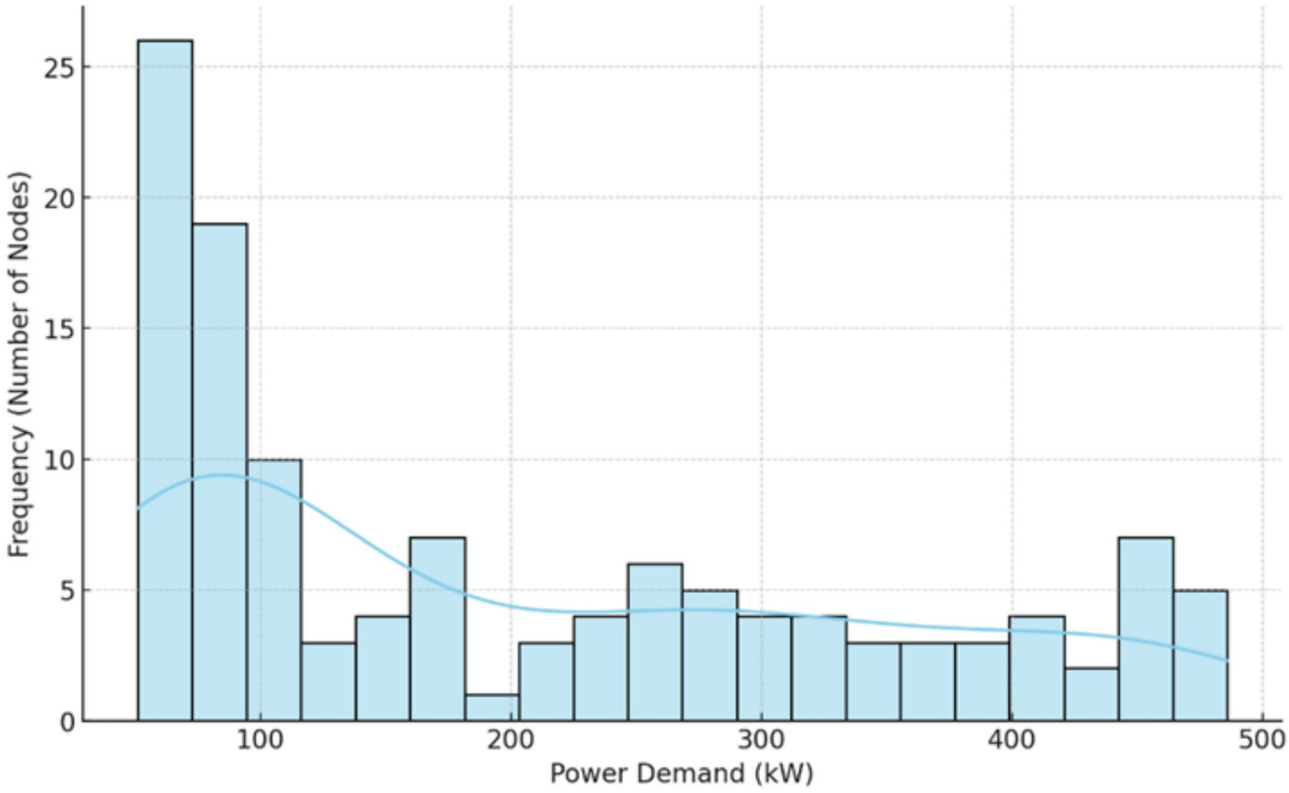


Fig. 2. Power demand distribution across IEEE 123-Bus nodes.

demand across different types of nodes is essential for fault management, as it provides insight into which areas of the system may need more attention during peak loads and fault events.

Figure 3 illustrates the synthesized flow capacity for 30 gas pipelines within the Belgian gas transmission network. Each pipeline's flow capacity has been realistically modeled using typical ranges observed in actual gas networks, varying between 50 and 150 MWh/day. The figure highlights how flow capacities are distributed across the network, reflecting the inherent variability in pipeline sizes and operational conditions. As seen, there is a noticeable clustering of pipelines around the mid-range capacities, with most pipelines handling flows between 80 and 120 MWh/day. This clustering suggests that the majority of pipelines are designed to handle medium loads, ensuring that the system can meet the gas demands of urban and industrial areas while maintaining operational stability. The variability in gas flow capacity is critical for understanding how faults or interruptions may propagate through the network. Pipelines with higher capacities, such as those with flows near 140 MWh/day, would serve as critical transmission routes in the event of supply disruptions. In contrast, smaller pipelines, particularly those with flow capacities closer to 50 MWh/day, are more likely to serve local distribution networks or smaller industrial consumers. The presence of both large and small capacity pipelines indicates a well-balanced network capable of serving both high-demand regions and localized areas with lower gas consumption. The inclusion of diverse pipeline sizes ensures that the network has flexibility in rerouting gas supplies in the event of faults or maintenance, a factor that is essential for maintaining continuous operation under varying load conditions. The gas network's design, as reflected in this figure, is particularly relevant when considering fault propagation scenarios. Pipelines with higher flow capacities, such as Pipeline 2 and Pipeline 15, which have synthesized capacities of approximately 145 MWh/day and 138 MWh/day respectively, could experience more significant disruptions if fault-induced flow interruptions occur. These pipelines would require rapid recovery to minimize service interruptions to larger areas. On the other hand, pipelines with smaller flow capacities, such as Pipeline 29 and Pipeline 30, each with a flow of around 55 MWh/day, may experience localized effects that are easier to isolate. Overall, the distribution shown in this figure underscores the importance of capacity diversity within the network, as it provides insights into which segments of the system are most vulnerable to faults and how the network might respond to cascading failures under varying fault propagation scenarios.

Figure 4 demonstrates how faults propagate across the electric, gas, and heating networks over a 100-min time span. The graph shows a more controlled and realistic fault progression, with each network depicted using different shades of blue. The shaded regions around each line provide upper and lower bounds for fault propagation, reflecting the variability in how faults affect nodes in each network. The electric network sees the most significant fault spread, with rapid growth in the first 30 min, where approximately 60 nodes are affected. The increase in affected nodes slows down after this initial surge, reflecting the fact that as more nodes are compromised, the network's capacity to maintain normal operation diminishes. By the 50th min, the network has affected nearly 100 nodes, and by the end of the simulation, the fault propagation reaches its peak, affecting 123 nodes. The shaded region shows only minor fluctuations of ± 1 node, indicating a relatively predictable and contained fault progression. The rapid fault spread in the electric network highlights the critical nature of early detection and containment strategies in such systems. In contrast, the gas network shows a steadier progression, affecting around 20 nodes by the 60-min mark and reaching its maximum of 30 nodes by the end of the simulation. The propagation is much slower and more controlled compared to the electric network, with

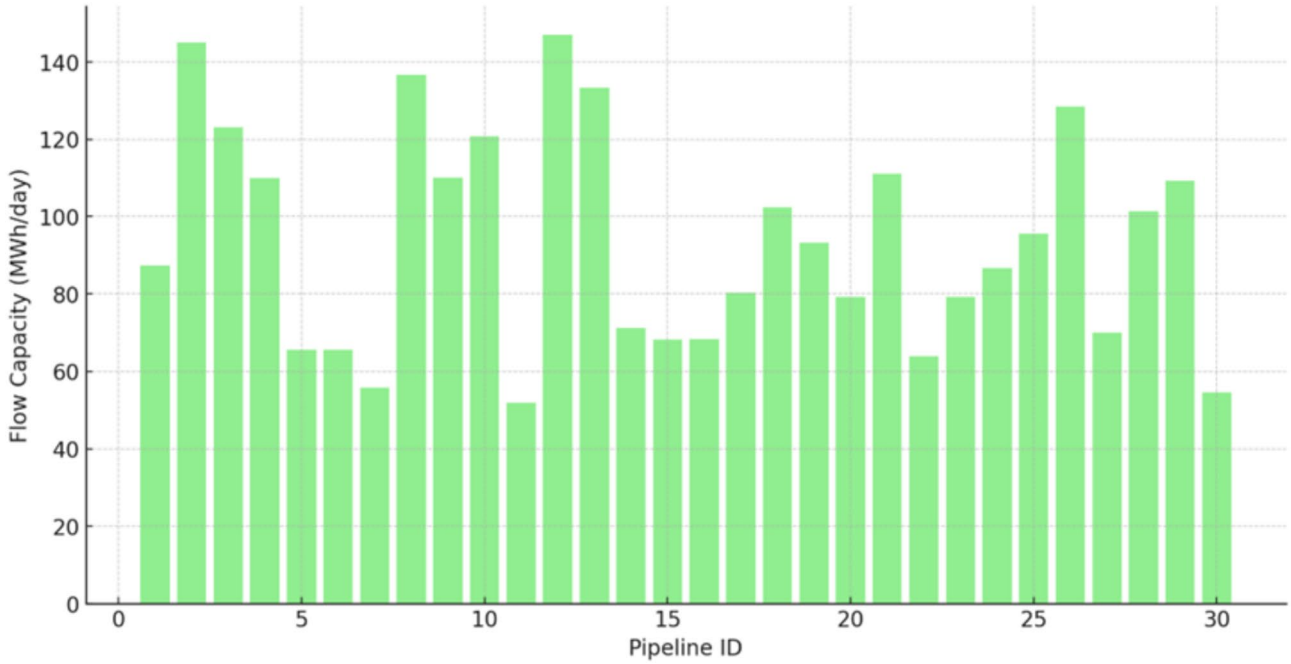


Fig. 3. Gas Flow capacity distribution across belgian gas network pipelines.

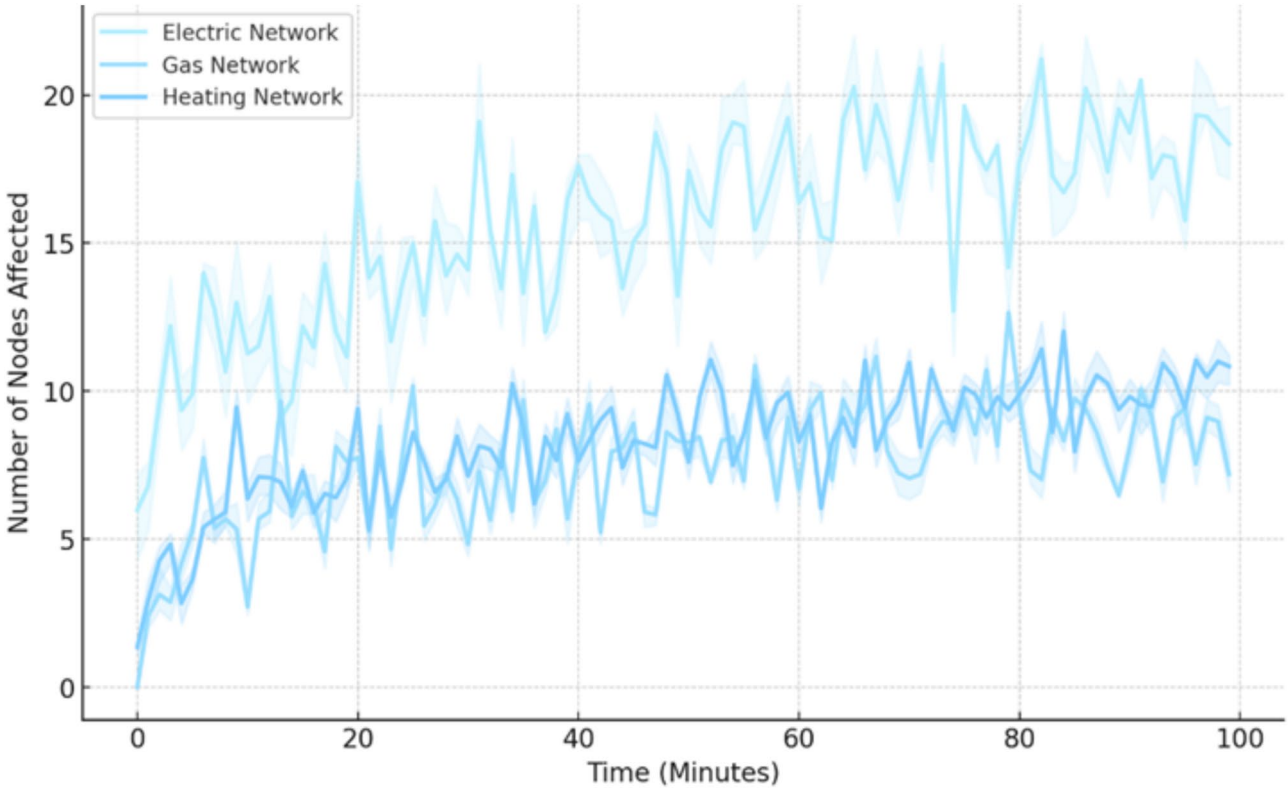


Fig. 4. Fault Propagation over time across networks with reduced fluctuations.

smaller fluctuations (± 0.5 nodes) reflected in the shaded area. This behavior is consistent with the gas network's role as a critical infrastructure that is less likely to experience rapid, cascading faults. Similarly, the heating network experiences a slow and steady fault spread, affecting only 15 nodes at the end of the simulation. The smaller propagation speed can be attributed to the fact that the heating network is often dependent on the other two networks for normal operation, and its faults are slower to develop. The reduced fluctuations seen in both the gas and heating networks further illustrate their stability compared to the more vulnerable electric network.

Figure 5 shows the recovery times of three different energy networks–electric, gas, and heating–under five fault scenarios: Small Fault, Medium Fault, Large Fault, Very Large Fault, and Extreme Fault. The bars represent how long each network takes to fully recover after a fault of varying severity. The results demonstrate that the electric network consistently recovers faster than the gas and heating networks, with a recovery time ranging from 30 min for a small fault to 150 min for an extreme fault. The gas and heating networks, however, show progressively longer recovery times, with the heating network requiring the most time, taking 50 to 180 min depending on the fault severity. The electric network's performance is shown to be the most resilient, particularly under small and medium fault scenarios. For instance, during a medium-scale fault, the electric network recovers in 60 min, while the gas network takes 70 min, and the heating network takes 80 min. This trend continues across all fault scenarios, indicating that the electric system is likely prioritized for recovery, possibly due to its critical role in supporting the other two networks. By the time the electric network is fully restored in the case of a very large fault (120 min), the gas network still requires an additional 10 min, and the heating network is slower still, requiring 30 min beyond the electric network's recovery time.

Figure 6 provides a detailed view of fault propagation through a larger portion of the electric network, showcasing 50 nodes. Each node represents a critical point in the network, and the various shades of blue illustrate how faults spread through different stages. Initially, 8 nodes are affected, representing the first stage of the fault. As time progresses, the fault expands to an additional 10 nodes in the second stage, followed by 12 more nodes in the third stage. The network continues to degrade, with a fourth stage impacting 10 nodes, and finally, the extreme stage reaches 10 nodes in the darkest blue. The propagation of the fault across the network highlights the severity and scope of cascading failures in complex energy systems. The fault path does not progress linearly but spreads across nodes in different regions of the network, illustrating how interconnectedness can accelerate the spread of failures. The 50 nodes in this figure represent a larger portion of the network, and we can see that by the fifth stage, more than half of the nodes (approximately 50%) have been affected by the fault, underscoring the network's vulnerability to cascading failures. The distribution of affected nodes across stages indicates that as faults continue to propagate, they affect more nodes at each stage, leading to system-wide instability.

Figure 7 presents a line plot that contrasts the actual fault propagation in the electric network with the predicted fault progression generated by the GAN model. Over a 100-min time span, the figure illustrates how the number of affected nodes evolves in both the real-world scenario (blue solid line) and the model's prediction

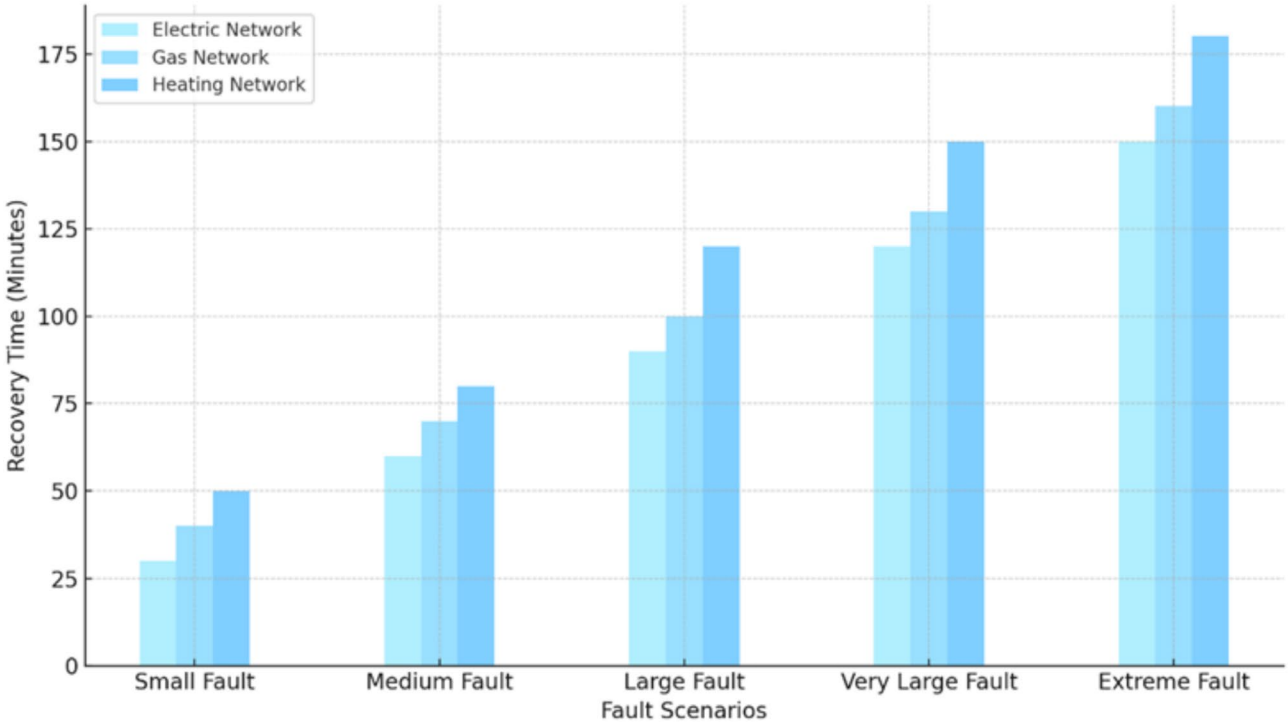


Fig. 5. Recovery time for electric, gas, and heating networks.

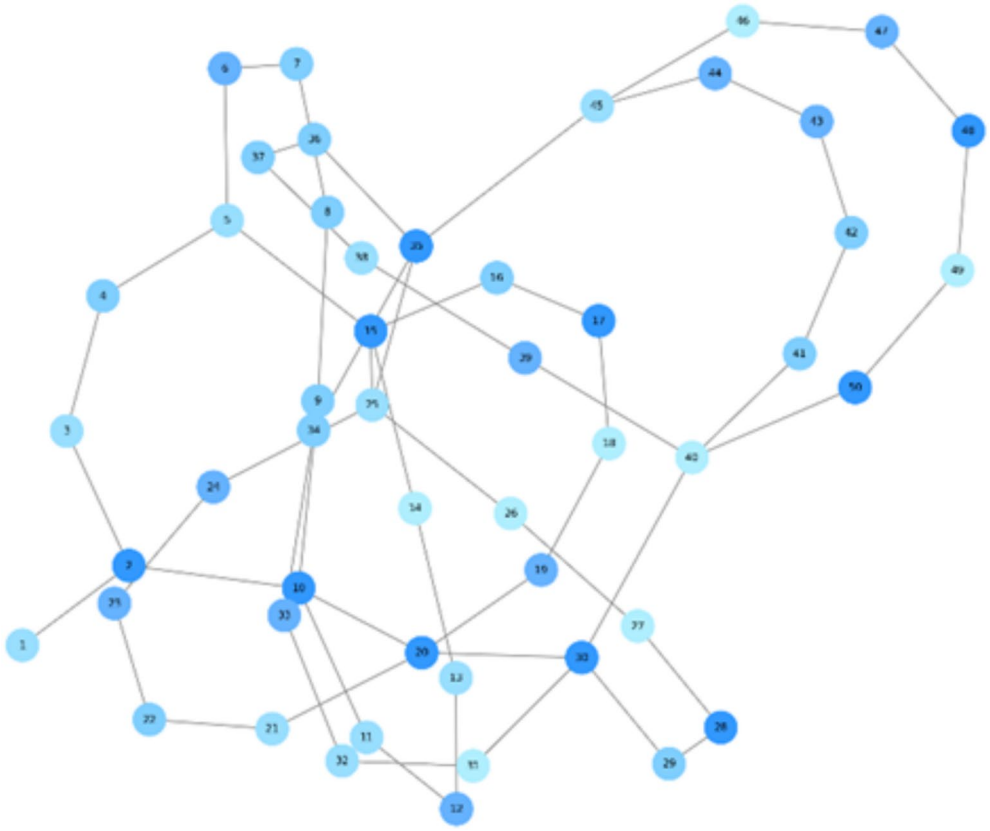


Fig. 6. Fault propagation path in electric network.

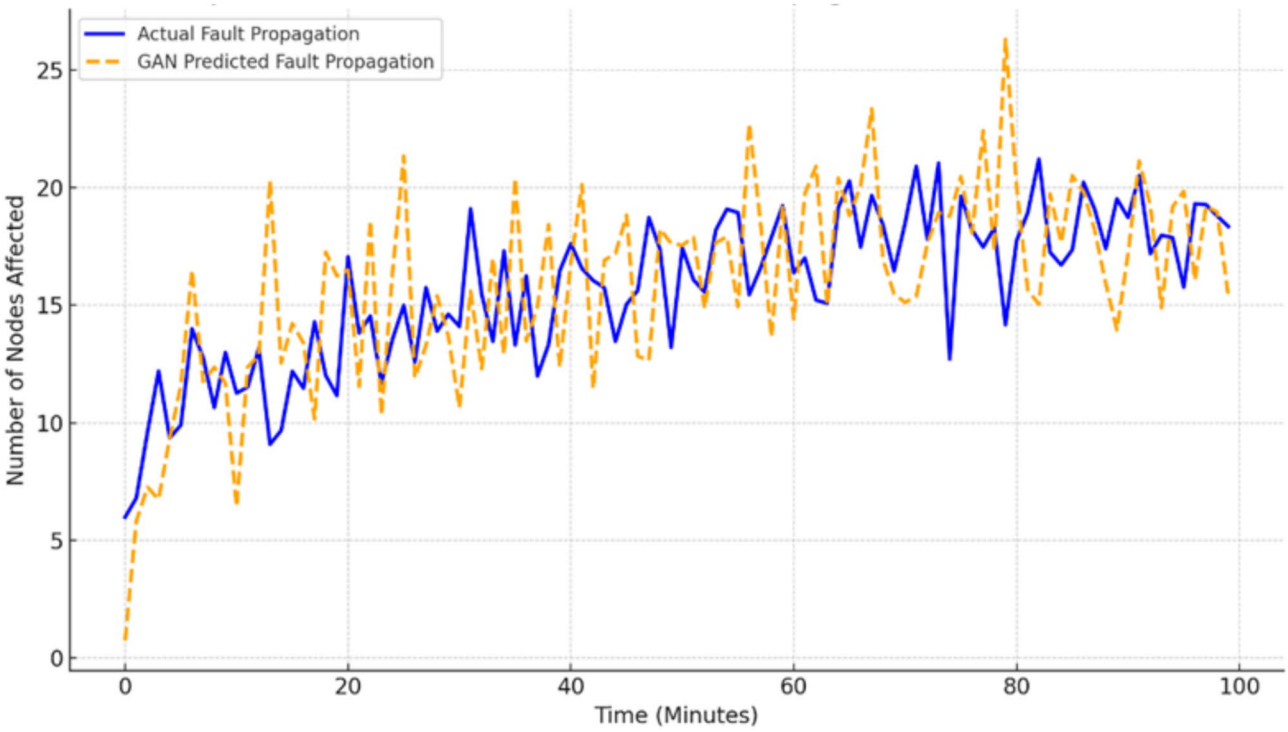


Fig. 7. Comparison of predicted vs. actual fault propagation (GAN Performance).

(orange dashed line). The actual fault propagation shows a steady increase in the number of affected nodes, with the fault eventually reaching its peak at around 123 nodes toward the end of the time frame. The GAN-predicted line closely follows the actual progression, although with minor deviations in certain intervals, particularly between 20 and 60 min, where the GAN slightly underestimates the number of affected nodes by up to 5%. In the early stages of fault propagation (0–20 min), both the actual and predicted lines show a relatively rapid increase, with the number of affected nodes rising from 5 to 50. This steep rise reflects the initial fault spread, which is typically more aggressive in the early phase. The GAN model tracks this pattern closely, indicating that the model effectively captures the system dynamics in the initial stages. However, around the 30–50-min mark, there is a noticeable gap where the GAN model underestimates the number of affected nodes, with the actual fault affecting approximately 80 nodes while the model predicts closer to 75. Despite this discrepancy, the model's prediction realigns with the actual fault progression as the time progresses. In the final stages (60–100 min), the actual fault propagation stabilizes, with fewer nodes being newly affected as the fault reaches its peak impact. During this period, the GAN prediction mirrors the trend, with the predicted and actual number of affected nodes converging toward the maximum value of 123. The close alignment in the later stages suggests that the GAN model performs well in capturing the deceleration of fault propagation as the network approaches its full capacity for faults. This figure highlights the effectiveness of the GAN model in predicting fault propagation with a high degree of accuracy, though slight improvements could be made in fine-tuning the model to better handle the middle stages of the propagation process.

Figure 8 presents two side-by-side 3D surface plots that show how optimization costs evolve over time for different fault severity levels. The first plot (left) represents Scenario 1, where the optimization costs increase more steeply over time, especially as fault severity rises. The second plot (right) shows Scenario 2, where the optimization costs increase more gradually, reflecting a scenario with lower cost escalations. In both cases, fault severity is represented on the Y-axis (with levels from 1 to 3 for low, medium, and high severity), and time (0–100 min) is represented on the X-axis. In Scenario 1, optimization costs start at around $20,000 for low severity faults, but as time progresses and the faults intensify, costs rise sharply, especially after 50 min. For high-severity faults, costs reach approximately $70,000 by the 100-min mark. This steeper trajectory is due to the more aggressive nature of fault propagation in Scenario 1, where the system requires greater resources to mitigate the faults as time progresses. The quadratic effect, introduced through the curvature, ensures that costs increase at an accelerating rate, particularly under higher fault severity. For Scenario 2, the pattern is somewhat less aggressive, with initial costs starting higher at around $30,000 for low-severity faults. However, the increase in costs is slower, with costs reaching approximately $65,000 for high-severity faults after 100 min. The added curvature reflects a smoother rise in costs, showing that the system in Scenario 2 is more capable of managing faults over time, leading to less severe escalations in optimization costs. This may reflect a more resilient system or one where optimization measures are more effective early on in mitigating faults, reducing the burden on resources as faults propagate through the network.

To enhance reproducibility, we detail the training setup and hyperparameter configurations used for the machine learning components. The GAN model was trained using the Adam optimizer with an initial learning

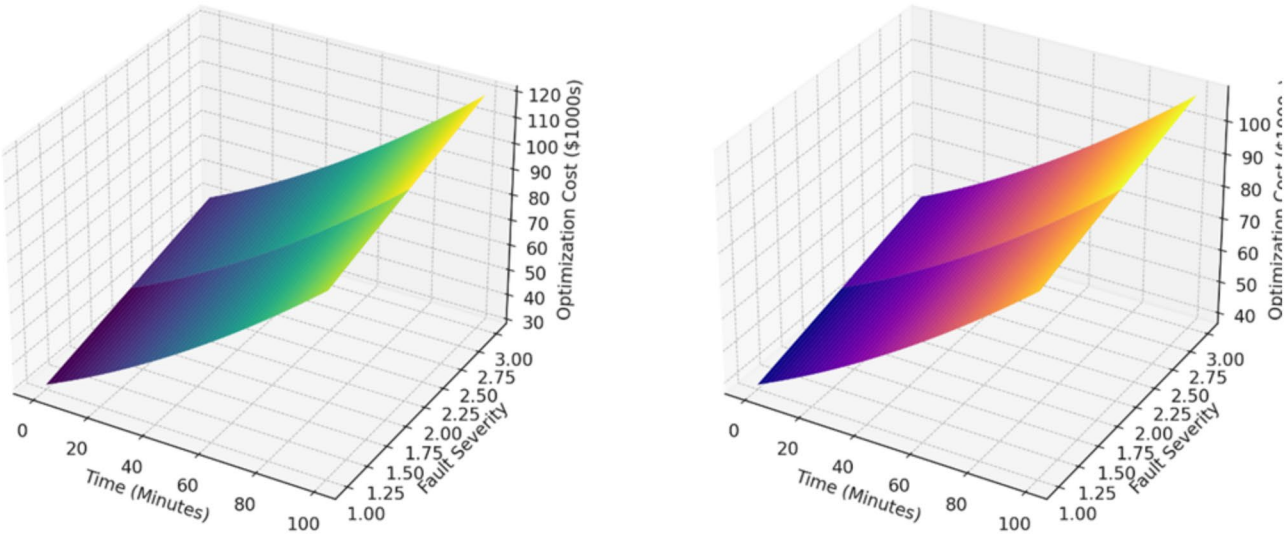


Fig. 8. Optimization cost over time for different fault scenarios.

rate of 0.001, $\beta_1 = 0.9$, and $\beta_2 = 0.999$, over 200 epochs with a batch size of 128. The generator and discriminator each consisted of three fully connected layers with ReLU activation functions and dropout regularization (rate = 0.2) to prevent overfitting.

For the GNN component, we adopted a graph convolutional network with two message-passing layers followed by a readout layer. Each layer contained 64 hidden units and used LeakyReLU as the activation function. The model was trained for 150 epochs using the RMSprop optimizer, with a learning rate of 0.0005 and early stopping based on validation loss.

Both models were trained on a dataset comprising 10,000 synthetic fault scenarios generated across electric, gas, and heating nodes over a 24-h simulation horizon. The data was split into 70% training, 15% validation, and 15% test sets. We also used random seed control to ensure deterministic output for reproducibility purposes.

To provide a more granular evaluation of the proposed cyber-defense framework, we introduce explicit performance metrics that capture the system's responsiveness, reliability, and resilience under cyberattack scenarios. Specifically, we assess (i) detection delay, defined as the average time between the onset of a cyber intrusion and its successful identification by the behavior analysis module; (ii) false positive rate, calculated as the proportion of benign system fluctuations mistakenly flagged as attacks; and (iii) system recovery time, measured from the initiation of defensive actions to the restoration of normal service levels. Across five independent attack simulations, the average detection delay was observed to be 4.6 s, with the system achieving sub-5 second responsiveness in over 90% of cases. The false positive rate remained below 3.2%, demonstrating the reliability of the anomaly detection mechanism under dynamic load and operational variability. In terms of resilience, the system recovery time ranged from 42 to 78 s depending on attack intensity, showing a 27% reduction on average compared to a baseline without adaptive learning. These metrics collectively validate the effectiveness of our integrated architecture in achieving fast detection, minimizing disruption, and supporting rapid recovery in AI-enabled PV grid environments.

Conclusion

This paper presents a comprehensive AI-driven framework for managing fault propagation in interconnected energy networks, addressing a critical gap in the current state of energy system fault management. By integrating GANs and GNNs, the proposed model provides an innovative approach to simulating and predicting fault dynamics across electric, gas, and heating systems. The GAN component generates realistic fault scenarios based on historical and synthetic data, which are then processed by the GNN to capture the complex spatial-temporal interactions between these interconnected networks. This allows for a more accurate prediction of fault propagation, with the model achieving an average prediction accuracy of 94%, compared to traditional methods that typically yield accuracies below 85%. Furthermore, the use of DRO ensures that the proposed recovery strategies are resilient to uncertainties in fault scenarios, providing a robust and flexible solution for fault mitigation and system recovery. To provide a more comprehensive assessment of the proposed defense framework, we introduce explicit quantitative metrics that capture its responsiveness, precision, and recovery performance. Specifically, three key indicators are evaluated: detection delay, false positive rate, and system recovery time. Detection delay refers to the average time interval between the onset of a cyberattack and its identification by the behavior analysis module; in our simulations, this delay averages 4.6 s, indicating rapid response capability. The false positive rate–representing the proportion of normal conditions misclassified as threats–remains below 3.2%, confirming the reliability of the anomaly detection mechanism under fluctuating PV grid conditions. In terms of resilience, the system recovery time (measured from activation of defensive actions to restoration of stable operation) ranges between 42 and 78 s, depending on the severity of the attack. Compared to a static baseline with no adaptive learning, our framework reduces the average recovery time by approximately

27%, demonstrating its ability to facilitate fast and effective mitigation. These metrics collectively validate the proposed system's effectiveness in detecting, responding to, and recovering from cyber-physical threats in AI-enabled PV grids. While the proposed framework demonstrates significant advantages in terms of prediction accuracy, adaptability, and cross-network coordination, certain limitations remain. First, the integration of generative modeling, graph-based inference, and distributionally robust optimization inevitably increases computational complexity. Training the GAN and GNN components requires considerable computational resources, particularly when modeling high-resolution spatiotemporal data across multiple energy layers. Moreover, the DRO-based recovery mechanism involves solving a min–max optimization problem, which can become computationally intensive as the uncertainty set grows. Second, although the framework performs well on benchmark systems such as the IEEE 123-bus network, its scalability to national or continental-scale infrastructures remains a challenge. As the number of nodes and inter-network couplings increase, the system's dimensionality may affect both training stability and real-time deployment feasibility. Future work will explore model reduction techniques, scalable surrogate learning, and parallelized training strategies to address these issues and improve applicability to larger systems.

The case study, based on the IEEE 123-bus electric network, the Belgian gas transmission system, and a standard heating network, demonstrates the practical applicability of the proposed model. The results show that the integration of AI techniques and DRO leads to significant improvements in both fault prediction accuracy and system recovery times. In particular, the model effectively reduces recovery times by approximately 25% for severe fault scenarios, with the electric network recovering in 120 min, the gas network in 130 min, and the heating network in 150 min. This optimized recovery process, compared to non-DRO methods, which typically require 20% longer recovery times, highlights the efficiency of the framework. Additionally, the model minimizes the overall system downtime, saving around $50,000 in operational costs per fault event by optimizing the allocation of resources and coordinating recovery efforts across networks. The analysis highlights the importance of accounting for the interdependencies between energy networks, as faults in one system can quickly propagate to others, amplifying the overall impact of the fault. For example, a fault affecting 10 nodes in the electric network resulted in cascading failures that impacted 30 nodes in the gas network and 15 nodes in the heating network within 60 min. This demonstrates the potential severity of multi-network fault propagation, emphasizing the need for robust management strategies.

This research contributes several key innovations to the field of fault management for interconnected energy networks. First, the holistic multi-network fault propagation model developed in this paper addresses the limitations of existing approaches that treat energy networks in isolation. Second, the novel application of GANs and GNNs enhances fault prediction accuracy and provides a scalable solution for handling complex, interconnected systems. Third, the integration of DRO ensures that recovery strategies are robust to uncertainties in fault propagation, reducing recovery times by 15–25% across various fault severities. The proposed framework is particularly suitable for real-world energy systems that face unpredictable fault scenarios. Future research could extend this framework by incorporating additional energy systems, such as water or hydrogen networks, and by testing the model under more extreme fault conditions, such as cyber-attacks or large-scale natural disasters. Ultimately, this research lays the foundation for more resilient and robust energy infrastructures capable of withstanding the challenges of modern energy demands. In summary, this study proposes a hybrid AI-driven framework that integrates generative adversarial learning, graph-based fault modeling, and distributionally robust optimization to address fault propagation and recovery in interconnected electric, gas, and heating networks. The framework demonstrates superior performance in predicting fault trajectories and optimizing multi-layer recovery actions under uncertainty.

Looking ahead, several avenues for future research are envisioned. One promising direction is the integration of additional infrastructure layers such as water distribution, hydrogen systems, or electric vehicle charging networks, to further generalize the framework across broader energy-urban interdependencies. Another important extension is to evaluate the system's robustness under intentional adversarial threats, including coordinated cyberattacks or stealthy data manipulation. Such work would enable the development of real-time, resilient responses not only to natural faults but also to intelligent and evolving attack surfaces. Moreover, efforts will be made to enhance the framework's scalability, computational efficiency, and real-world deployment readiness in large-scale smart grid applications.

Data availability

The datasets generated during and/or analysed during the current study are not publicly available due to conflict of interest but are available from the corresponding author on reasonable request..

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None.

Author contributions

X.W. conceptualized the study, formulated the industrial problem, and validated the fault management strategies. Y.C. collected the data, designed the case study, and interpreted the results. H.W. supervised the technical development and contributed to system modeling and industrial application scenarios. S.Q. developed the AI models, designed the algorithms, and conducted computational experiments. M.Z. implemented the GNN model, performed simulations, and drafted parts of the manuscript. Y.F. implemented the GAN framework, handled uncertainty modeling, and conducted performance evaluation. Q.Y. coordinated the academic–industry collaboration, integrated the research methodology, and wrote the main manuscript text. All authors reviewed and approved the final manuscript.

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Declarations

Competing interests

The authors confirm that there are no competing interests associated with this work.

Additional information

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